



Instructor: Date:

Learner: Regional Branch:

Program: Center:

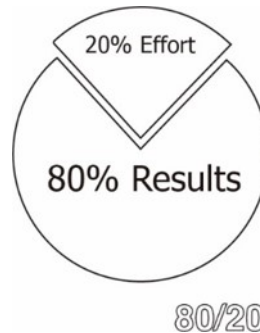
CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

We need to treat the process of defect tracking wisely.



Learning Outcome:

Describe ways of solving problems and getting things done in written and spoken situations and in work and social contexts.

Train the system users on application structuring and management, according to the established plan and user profile, according to the organization policies.

Ask your partner:

- Do you think that all software defects are equally important and should be tracked under the same parameters? Why?
- How do you collect data for defect tracking processes?

Can Do Statements:

- I can say why the principle of collecting more defects may not work in some cases.
- I can list some questions an organization should ask to decide when to start tracking defects.
- I can talk about the relation of the Pareto Principle to defect tracking.



Let's Explore!

Let's pinpoint* the source of the most relevant defects.

* We use the verb "*pinpoint*" for expressing that we are going to identify with precision the location of something.



Instructions: Read the text below.

Sophia: So, the software development team was lagging behind because they were tracking defects?

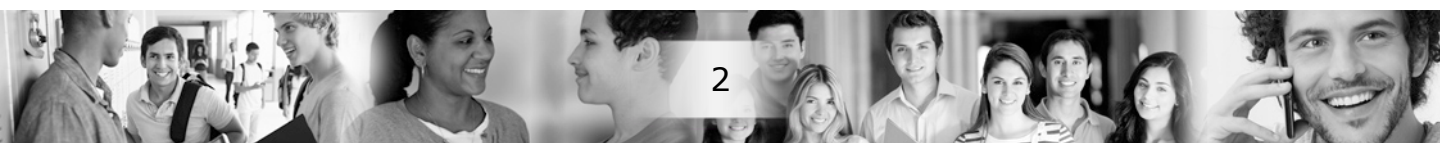
Charlee: Not exactly. Every organization should have their defects tracked in order to get them fixed and to help find patterns of where most defects are coming from. However, our greatest trouble was having unnecessary data entered by the members of the team; thus, they were losing too much time when tracking defects to the point where this process became ineffective.

Sophia: But I thought having detailed information entered could enrich the database and facilitate the fixing process.

Charlee: Up to a certain point. As they were spending too much time doing this and were not able to work to their full potential in other processes, they started to overlook some defects to save time. Therefore, some important defects were passing unnoticed.

For this reason, in order to know when tracking defects is truly necessary, I suggest having some questions addressed to get to know more about how this process should be carried out. These question are: What are the most common types of defects that are causing most of the problems in software quality? When are these defects injected into the product? Who is going to use the collected data for tracked defects? And how is this data going to help improve the product?

These questions respond to the Pareto Principle in which most likely, most defects are found in one or two processes. As we need to be effective and aim to improve quality, we must have the defects that affect the software development process the most tracked with more attention and detail. If every organization has their most relevant defects prioritized, they can improve their defect tracking process just as the Pareto Principle suggests.



**Let's Practice!**

Does it help to defect tracking?

Instructions: Match column A with column B.

COLUMN A

1. Every organization should
2. If every organization gets their most relevant defects prioritized,
3. Every organization
4. We must have the defects
5. I suggest every organization

COLUMN B

- a. they can improve their defect tracking process just as the Pareto Principle suggests.
- b. tracked with more attention and detail.
- c. should get their defects fixed.
- d. have their defects tracked.
- e. having these essential questions addressed to get to know more about how this process should be carried out.

**Let's Practice!**

Is it really useful?

Instructions: Select the correct answer.

1. The company _____ defects analyzed.
a. gets its b. get it c. get its d. got it
2. The employees _____ software examined.
a. has their b. have their c. have them d. has them
3. This data _____ improved.
a. have defects b. have the defects c. has the defects d. have a defect
4. Pareto Principle _____ tracking process upgraded.
a. have defects b. has defects c. have the defects d. has the defect
5. Designers should _____ defects tracked.
a. get their b. got them c. get them d. got their





Extension Activity

Let's research about the Pareto Principle.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about the Pareto Principle.
- Research about different examples of it in Software development.
- Make a list of examples where software developers use the Pareto Principle.
- Create a chart where you explain the scenario and the use of the Pareto Principle.
- Share your chart with the class.

Self-Assessment

- | | Yes | No |
|--|-----------------------|-----------------------|
| 1. I look for relevant information about the Pareto Principle. | ☐ | ☐ |
| 2. I explain the Pareto Principle in software development. | ☐ | ☐ |
| 3. I make a list of examples where software developers use the Pareto Principle. | ☐ | ☐ |
| 4. I create a chart to exemplify the Pareto Principle. | ☐ | ☐ |
| 5. I share my work with the class. | ☐ | ☐ |



Target Words

- inject defects into a product** (expression): to originate defects in a product at a moment of the development process.
e.g. When are these defects injected into the product?

- facilitate** (verb): to make something easier.
e.g. I thought having detailed information entered could enrich the database and facilitate the fixing process.

- overlook** (verb): to ignore, neglect.
e.g. they started to overlook some defects to save time. Therefore, some important defects were passing unnoticed.

- lag behind** (phrasal verb): to stay behind, not to be at expected or required level.
e.g. So, the software development team was lagging behind because they were tracking defects?

- enrich** (verb): to add more value or detailed information to something.
e.g. I thought having detailed information entered could enrich the database and facilitate the fixing process.

- pass unnoticed** (expression): to be ignored or not noticed.
e.g. Some important defects were passing unnoticed.



Learning Points

Causative Get and Have

We use **causatives** when we want to express that a person is causing something to happen.

We can use the verb **get** or the verb **have** when we want to express the idea of getting something done.

To use **causative verbs**, we are going to follow the next structure:

Causative Verb (**Get / Have**) + Object + **Verb in Past Participle**

Example:

*The manager is going to **have** the defects properly **tracked**.*

*I **got** the data **collected** last week.*

Examples

- *Every organization should have their defects tracked in order to get them fixed.*
- *I suggest having some questions addressed to get to know more about how this process should be carried out.*

Understanding the Pareto Principle

The Pareto Principle states that most things in life are not distributed evenly.

This principle originally referred to the fact that 80% of Italy's wealth belonged to the 20% of the population. Nevertheless, this principle can be applied to many other situations in life. For instance, in software development, we need to take into account that not all the bugs are going to cause the same damage to the software. Most likely, around 20% of the bugs are going to be the ones responsible for causing crashes, and the remaining 80% will either cause minor imperfections or may even pass unnoticed.

For example, only a 10% of the total applications developed in a year could have a considerable amount of downloads or the first 20% of your time reading an article is going to get you through the most important concepts. Knowing this can be very useful; once we identify the portion where the most effort, reward, or output is located, we will know where to focus most of our attention on.

Once an organization finds that the 20% percent of the defects that are causing the software to crash are originated in the coding process, then, this organization can make sure to pay more attention to this area and improve its processes.

